

Qns

2(a)(i) $u_y = u \sin \theta$

$$= 20 \sin 45^\circ$$

$$= 14 \text{ ms}^{-1}$$

2(a)(ii) $v_y^2 = u_y^2 + 2a_y s_y$

$$0 = (20 \sin 45^\circ)^2 + 2(-9.81)s_y$$

$$s_y = 10 \text{ m (shown)}$$

2(a)(iii)1.

$$K = \frac{1}{2}mu^2$$

At 10 m,

$$E_{k10} = \frac{1}{2}mv_x^2$$

$$= \frac{1}{2}mu_x^2 \quad (\text{because } v_x = u_x)$$

$$= \frac{1}{2}m(u \cos 45^\circ)^2$$

$$= \frac{1}{2}mu^2 \left(\frac{1}{2} \right)$$

$$= \frac{1}{2}K$$

$$= 0.5K$$

total energy = initial KE = K

By conservation of energy,

$$E_{p10} = \text{total energy} - E_{k10}$$

$$= K - \frac{1}{2}K$$

$$= \frac{1}{2}K$$

$$= 0.5K$$

2(a)(iii)2. At 10 m,

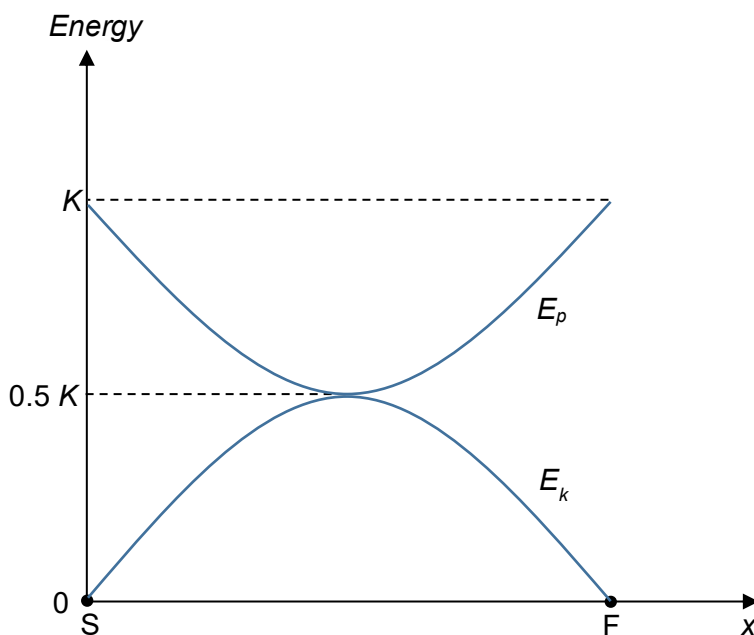
$$\begin{aligned} E_{p10} &= mg(10) \\ &= \frac{1}{2}K \quad (\text{from part 1}) \end{aligned}$$

At 5.0 m,

$$\begin{aligned} E_{p5} &= mg(5.0) \\ &= \frac{1}{2}mg(10) \\ &= \frac{1}{2}\left(\frac{1}{2}K\right) \\ &= \frac{1}{4}K \\ &= 0.25K \end{aligned}$$

$$\begin{aligned} E_{p5} &= \text{total energy} - E_{k5} \\ &= K - \frac{1}{4}K \\ &= \frac{3}{4}K \\ &= 0.75K \end{aligned}$$

2(b)



(i) shape correct (parabolic and symmetrical about centre of SF)
 $E_p = 0.5 K$ at maximum point clearly shown and labelled

(ii) shape correct (parabolic and symmetrical about centre of SF)
 $E_k = 0.5 K$ at minimum point clearly shown and labelled
 $E_k = K$ at S and F clearly shown and labelled

3(a) work done per unit mass in moving small test mass from infinity to the point in the gravitational field